AI BUILDER CUP 2026 · JAPAC

Training Failure Triage

11 / 14

A training job fails. The task is to decide **whether the fault is in the software or in the card**, and to show the evidence the decision rests on.

Detect | Collect evidence | Diagnose & plan | Act · notify · verify

11 / 14 above = the rule baseline's score on the 14-case library (chapter 03)

gpu-waste-auditor-2026.web.app

Sustainability & Social Impact

Three chapters

Each chapter ends on a number you can rerun.

A benchmark score, a tool-call count, an operation log. Where a number was never measured, the page shows a blank instead of a plausible value.

detect → collect evidence → diagnose & plan → act, notify, verify

01 **The problem** — One error code, two opposite readings, both expensive — 03 — 05

02 **What we built** — The agent reads the cluster, decides, acts, then reads back — 06 — 10

03 **Evidence & boundaries** — The rule baseline scores 11 of 14; the 3 misses are listed — 11 — 14

CHAPTER 01

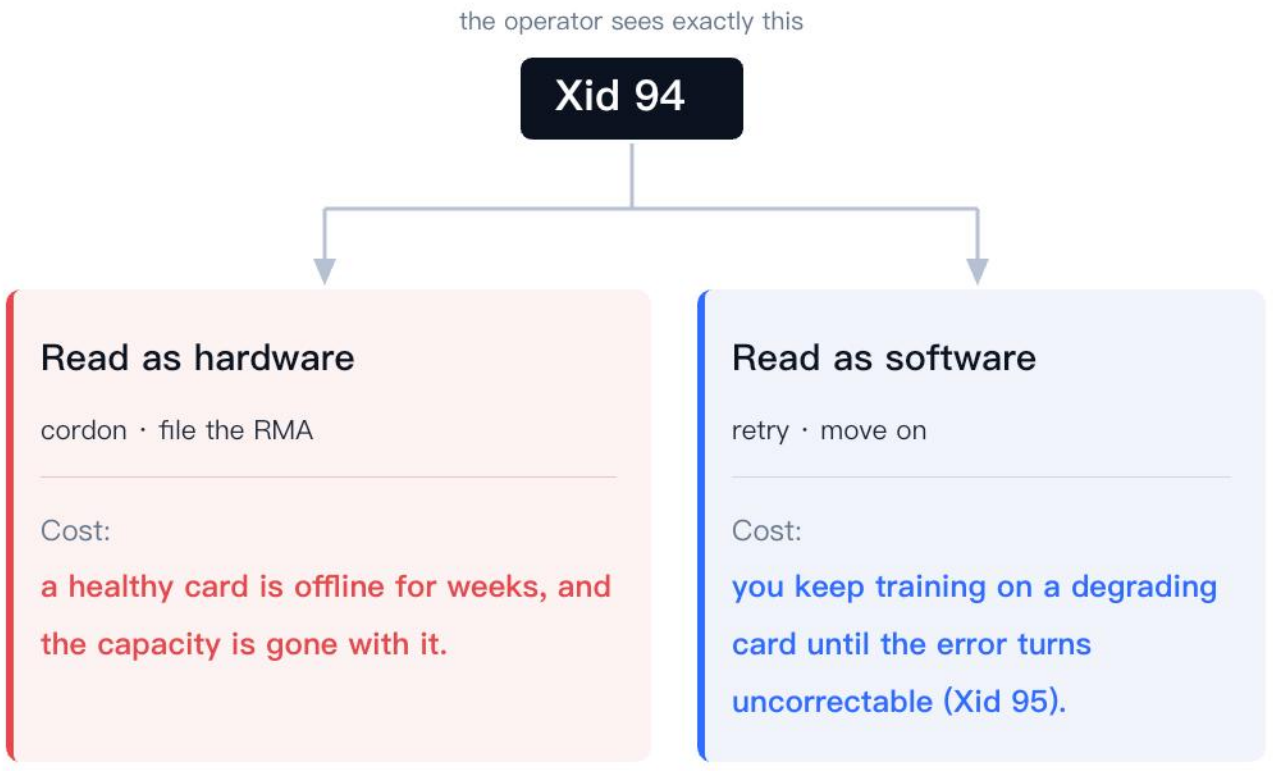
The problem

The same error code, two opposite mistakes

94

Calling a dying card healthy burns weeks of compute on it. Calling a healthy card dying pulls a working GPU out of the pool. Both are expensive, and telling the two apart is the whole problem.

One code, two opposite mistakes



Why the code cannot be the rule

- Xid 94 is a correctable ECC error. It tells you something went wrong; it does not tell you what to do.
- What separates the two nodes is how often the code fired in 30 days, and whether correctable ECC is rising at the same time.
- A lookup table maps one code to one verdict. Comparing a count against a trend needs something that can hold two numbers at once.

The code alone cannot choose between them. The 30-day trend can.

Same code. Opposite conclusions.

lyg2103 vs lyg2112 · 30-day window

1

lyg2103

Xid 94 in the last 30 days

1 correctable ECC event, flat

hardware_watch · no RMA

53

lyg2112

Xid 94 in the last 30 days

118 correctable ECC events, still rising

hardware · RMA filed

lyg2103	H800_80G · healthy Xid 94（已隔离的内存错误（Contained））×1，GPU 5，最近 2026-10-14T01:22:00Z 近 30 天 Xid 94 累计 1 次 可纠正 ECC 当前计数 1	任务：job-7741(htli)

two rows in between (lyg2107, lyg2108), omitted — both rows above are from the same list

lyg2112	H800_80G · healthy Xid 94（已隔离的内存错误（Contained））×41，GPU 3，最近 2026-10-14T03:58:00Z Xid 94（已隔离的内存错误（Contained））×12，GPU 3，最近 2026-10-13T21:10:00Z 近 30 天 Xid 94 累计 53 次 可纠正 ECC 呈 rising 趋势 可纠正 ECC 当前计数 118	任务：job-7903(chenl)

Both nodes reported Xid 94. What separated them was how often it fired, and whether the count was still climbing.

Both rows read live from the cluster tab, over the same 30-day window

CHAPTER 02

What we built

An agent that runs a whole shift from a single click

17

The shift recorded for our demo video made 17 tool calls: one cluster scan, six node reads, the knowledge-base lookups, then the write, the ticket and the notification. The loop is capped at 12 turns, so a shift that runs long is continued from the calls already made instead of restarted.

No log was pasted in



The cluster tab as it renders on load. The header reads: 6 nodes, 4 of them showing an anomaly, read live from Cloud Firestore. The interface is in Chinese because the on-call team is Chinese-speaking; every reading used in this deck is translated on the right.

Cluster tab · read live from Firestore

Where those numbers come from

6 nodes in the demo cluster

4 of them showing an anomaly right now

0 log lines pasted in by the operator

Every figure above is read from Cloud Firestore when the page loads, and the header says so. The shift itself starts from a button: this build is a static site with no resident process, so a real deployment would have a scheduler fire it.

Captured from the deployed page at gpu-waste-auditor-2026.web.app

The writes actually landed

Operation log · write, then re-read



2 / 2

actions executed, then re-read from the server to confirm

lyg2112.cordoned	false	→ **true**
lyg2115.cordoned	false	→ **true**

A write is only logged as verified after the app reads the field back from Firestore.

A write that returns success but leaves the value unchanged is logged as “verification failed” instead. Open the cluster collection in the Firebase console during a run and the two states move while you watch.

Both cordons are low-risk actions, so the agent was allowed to run them itself

The model does not set the risk level

LOW	6	Task-local, reversible, never touches hardware or the system. the agent may run it itself
MEDIUM	5	Affects other jobs on the same node, or needs a short interruption. per-item approval + impact acknowledgement
HIGH	3	Irreversible, touches hardware assets, or crosses nodes. approval + change ticket + second confirmation

A fifteenth action is registered and deliberately withheld: raise_nccl_timeout. Raising the timeout would suppress the symptom instead of fixing it, so the knowledge base marks it discouraged.

14 actions · 6 low / 5 medium / 3 high

The gate is a whitelist, not a prompt rule

```
if (act.risk !== 'low') {
  return { refused: true }
}
```

The tool schema has no risk field, so the model has no way to call an action low-risk.

A refused action is stopped inside the tool.



The three levels above are the page's own header, captured as it renders

Where every number on this page comes from

case-xid94-rising · single triage

hardware **RMA filed** **0.95**

model confidence

4 tool calls before it was allowed to answer

expected: hardware · RMA

AI tool-call trace: arguments printed exactly as they were sent

lookup_xid	{"code":94}
lookup_xid	{"code":95}
inspect_node	{"node":"lyg2103"}
propose_fix	{"action_id":"file_rma","blocks_job":true,"should_rma":true,"detail":"节点 lyg2103 上的 GPU 硬件劣化，两周内 Xid 94 频发（共 7 次）并已扩散为未隔离的 Xid 95 错误，ECC 不可纠正计数上升，建议立即提报 RMA 并下线该卡。","evidence":"Xid 94 历史频发，伴随不可纠正 ECC 和未隔离的 Xid 95 错误。","title":"GPU 硬件劣化提报 RMA","evidence_refs":["T1","T2","T3"],"confidence":0.95,"classification":"hardware"}
引用表	T1 = lookup_xid T2 = lookup_xid T3 = inspect_node

Numbers come from tools, not from the model

The model is not allowed to compute or recall a number. lookup_xid reads the Xid knowledge base, inspect_node reads node telemetry. A value that was never fetched cannot appear in the conclusion.

T1 · T2 · T3

Citation ids are attached by the app when a call actually executes. A reference the model invents is dropped, and the model is told it was dropped.

CHAPTER 03

Evidence & boundaries

Reproducible numbers, stated limits

Every number in this chapter can be rerun from a script in the repository. Where we never ran a case, the table shows a blank rather than a plausible value.

11 / 14

rule baseline on the 14–case library

11 / 14, and you can rerun it

Rule baseline · offline · zero model calls

11 / 14

78.6% · classification only

The three it gets wrong

case-xid48-dbe

a rising correctable-ECC trend read as an application bug

case-xid94-rising

the pair that needs the 30-day window, not the code

case-gpu-hang-deadlock

a secondary hang taken for the root cause

The baseline does two things only: vote over pre-analysis hits, and "hardware means file an RMA".

案例	期望分类	期望报修	规则给出	规则结果	AI 给出	AI 结果
case-oom-batch	config	否	config · 不报修	对	未跑	—
case-oom-fragmentation	config	否	config · 不报修	对	未跑	—
case-nccl-nvlink	network	否	network · 不报修	对	未跑	—
case-illegal-access	application	否	application · 不报修	对	未跑	—
case-driver-old	environment	否	environment · 不报修	对	未跑	—
case-xid48-dbe	hardware	是	application · 不报修	错	未跑	—
case-xid94-single	hardware_watch	否	hardware_watch · 不报修	对	未跑	—
case-xid94-rising	hardware	是	hardware_watch · 不报修	错	未跑	—
case-xid79-bus	hardware	是	hardware · 报修	对	未跑	—
case-gsp-119	environment	否	environment · 不报修	对	未跑	—
case-driver-nvml-mismatch	environment	否	environment · 不报修	对	未跑	—
case-smi-driver-dead	hardware	是	hardware · 报修	对	未跑	—
case-gpu-hang-xid62	environment	否	environment · 不报修	对	未跑	—
case-gpu-hang-deadlock	application	否	network · 不报修	错	未跑	—

Same case library, same scoring script. The AI column is blank wherever we did not run the AI, because a blank is the honest value there.

Reproduced by scripts/verify_triage.mjs — the page and the script share one code path

Five layers, one rule

01 **Knowledge base**

24 Xid entries · 27 failure signatures · 6 diagnostic principles

02 **Deterministic pre-analysis**

version stack, memory accounting, structured facts — no model involved

03 **Tools**

10 registered · 6 exposed in single triage · 10 in shift patrol

04 **Model loop**

Gemini · at most 12 tool turns · calls already made are carried over if the model is swapped mid-shift

05 **Execution**

low risk only, then the value is re-read and logged

Architecture

One rule, enforced twice

Numbers must come from a tool. Risk levels must come from the action library. The model is not allowed to declare either one.

Both rules are written in the prompt and re-checked in code. The prompt exists so the model understands the system; the code is what makes the rule hold.

Prompt rules can be argued with.

Code cannot.

A mode is a capability boundary: the tool list is cut to the mode before the loop starts

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Acted on, not just recommended

The demo run ends with a verdict, an operation log entry, and a read-back that proves the write landed.

live demo

gpu-waste-auditor-2026.web.app

rerun the numbers

verify_triage.mjs rule baseline and result invariants

verify_pipeline.mjs 81 / 81, fake transport and in-memory cluster

browser_check.mjs 31 / 31, real Chrome against real Firestore